Subject – FSD

Experiment No. 1

PROFILE.HTML –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Profile</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
  <h1>My Profile</h1>
  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Contact</a>
  </nav>
</header>

<!-- Profile Section -->
<section class="profile-container">
  <div class="profile-left">
    <img src="C:\Users\Astik\Downloads\profilepic.jpeg" alt="Profile Image">
  </div>
  <div class="profile-right">
    <h2>Astik Pawar</h2>

    <p class="bio">
    I am a passionate and dedicated Computer Science student with a strong interest in
    Full Stack Development and Cloud Computing. I enjoy building responsive and scalable
    web applications using modern technologies. I am always eager to learn new tools,
    improve my problem-solving skills, and work on innovative projects that create
    real-world impact.
    </p>

    <p><strong>Email:</strong> astikpawar007@gmail.com</p>
    <p><strong>Phone:</strong> +91 9322809650</p>
  </div>

</section>

<!-- Technical Skills Section -->
<section class="technical">
  <h2 class="section-title">Technical Skills</h2>
```

```
    <div class="skills">

        <div class="skill-card">
            <h3>Python</h3>
            <p>Data analysis and backend development</p>
        </div>
        <div class="skill-card">
            <h3>Cloud Computing</h3>
            <p>AWS</p>
        </div>
        <div class="skill-card">
            <h3>Machine Learning</h3>
            <p>Building predictive models</p>
        </div>
        <div class="skill-card">
            <h3>Git & GitHub</h3>
            <p>Version control and collaboration</p>
        </div>

    </div>
</section>
<footer>
    <p>© 2026 Astik Pawar. All rights reserved.</p>
</footer>
```

STYLE.CSS –

```
/* General Styling */
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: linear-gradient(to right, #4e73df, #1cc88a);
}
/* Header */
header {
    background-color: #222;
    color: white;
    text-align: center;
    padding: 20px;
}
nav {
    margin-top: 10px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 15px;
    font-weight: bold;
```

```
    transition: color 0.3s;
}
nav a:hover {
    color: #1cc88a;
}
.profile-container {
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 60px;
    background-color: white;
    margin: 40px auto;
    padding: 40px;
    width: 75%;
    border-radius: 12px;
    box-shadow: 0 6px 15px rgba(0,0,0,0.3);
}
.profile-left img {
    width: 220px;
    height: 220px;
    border-radius: 50%;
    object-fit: cover;
}
.profile-right {
    max-width: 600px;
}
.profile-right h2 {
    margin-bottom: 15px;
}
.bio {
    line-height: 1.6;
    color: #555;
    margin-bottom: 15px;
}

/* ===== SKILLS SECTION ===== */

.section-title {
    text-align: center;
    margin-top: 40px;
    font-size: 28px;
    color: white;
}
.technical {
    padding-bottom: 40px;
}

.skills {
    display: flex;
    justify-content: center;
```

```
    gap: 25px;
    padding: 40px;
    flex-wrap: wrap;
}
.skill-card {
    background-color: white;
    width: 220px;
    padding: 20px;
    text-align: center;
    border-radius: 10px;
    box-shadow: 0 4px 10px rgba(0,0,0,0.2);
    transition: transform 0.3s, box-shadow 0.3s;
}
.skill-card:hover {
    transform: translateY(-8px);
    box-shadow: 0 8px 18px rgba(0,0,0,0.3);
}

/* Footer */
footer {
    background-color: #222;
    color: white;
    text-align: center;
    padding: 15px;
}
/* ===== RESPONSIVE DESIGN ===== */

@media (max-width: 768px) {

    nav a {
        display: block;
        margin: 10px 0;
    }
    .profile-container {
        flex-direction: column;
        text-align: center;

    .skills {
        flex-direction: column;
        align-items: center;
    }
    .profile-container {
        width: 90%;
    }
}
```

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Experiment No. 2

Json.html -

```
<!DOCTYPE html>
<html>
<head>
  <title>JSON</title>
</head>
<body style="font-family: Arial; background-color: #f4f6f9; margin: 0;">

  <div style="background-color: #2c3e50; color: white; padding: 10px 20px;">
    <h3 style="margin: 0;">Experiment No. 2</h3>
  </div>

  <div style="padding: 20px;">

    <h1 style="color: #2c3e50;">Student JSON Program</h1>

    <h2>All Students (JSON Format)</h2>
    <pre id="jsonOutput"
      style="background: #ffffff; padding: 15px; border: 1px solid #ccc;">
    </pre>

    <h2>Passed Students (Marks > 40)</h2>
    <pre id="passedOutput"
      style="background: #ffffff; padding: 15px; border: 1px solid #ccc;">
    </pre>

  </div>

<script>

class Student {
  constructor(rollNo, name, marks) {
    this.rollNo = rollNo;
    this.name = name;
    this.marks = marks;
  }

  displayInfo() {
    return `Roll No: ${this.rollNo}, Name: ${this.name}, Marks: ${this.marks}`;
  }
}

// Creating array of 5 students
let students = [
  new Student(1, "Astik", 75),
  new Student(2, "Rahul", 35),
  new Student(3, "Sneha", 60),
  new Student(4, "Priya", 42),
  new Student(5, "Amit", 28)
```

```
];

// Convert to JSON
let jsonData = JSON.stringify(students, null, 4);
document.getElementById("jsonOutput").textContent = jsonData;

// Function to get passed students
function getPassedStudents() {
    return students.filter(student => student.marks > 40);
}

let passedStudents = getPassedStudents();
document.getElementById("passedOutput").textContent =
    JSON.stringify(passedStudents, null, 4);

</script>

</body>
</html>
```

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Experiment No. 3

LambdaOperation.java –

```
import java.util.Scanner;

@FunctionalInterface
interface Operation {
   int compute(int a, int b);
}

public class LambdaOperations {
   public static void main(String[] args) {

      Scanner sc = new Scanner(System.in);

      System.out.print("Enter first number: ");
      int a = sc.nextInt();

      System.out.print("Enter second number: ");
      int b = sc.nextInt();

      // Lambda expressions
      Operation add = (x, y) -> x + y;
      Operation sub = (x, y) -> x - y;
      Operation mul = (x, y) -> x * y;
      Operation div = (x, y) -> y != 0 ? x / y : 0;

      System.out.println("Addition: " + add.compute(a, b));
      System.out.println("Subtraction: " + sub.compute(a, b));
      System.out.println("Multiplication: " + mul.compute(a, b));
      System.out.println("Division: " + div.compute(a, b));

      sc.close();
   }
}
```

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Experiment No. 4

StudentCRUD.java –

```
import java.sql.*;

public class StudentsCRUD {

    static final String URL = "jdbc:mysql://localhost:3306/college";
    static final String USER = "root";
    static final String PASSWORD = "astik45";

    public static void main(String[] args) {

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(URL, USER, PASSWORD);

            // INSERT
            String insertQuery = "INSERT INTO Students VALUES (1, 'Astik', 85)";
            Statement stmt = con.createStatement();
            stmt.executeUpdate(insertQuery);
            System.out.println("Record Inserted");

            // SELECT
            ResultSet rs = stmt.executeQuery("SELECT * FROM Students");
            System.out.println("\nStudent Records:");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("rollNo") + " " +
                    rs.getString("name") + " " +
                    rs.getInt("marks")
                );
            }

            // UPDATE
            String updateQuery = "UPDATE Students SET marks=90 WHERE rollNo=1";
            stmt.executeUpdate(updateQuery);
            System.out.println("\nRecord Updated");

            // DELETE
            String deleteQuery = "DELETE FROM Students WHERE rollNo=1";
            stmt.executeUpdate(deleteQuery);
            System.out.println("Record Deleted");

            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

// run the commands into terminal

// javac -cp ".;mysql-connector-j-9.6.0.jar" StudentsCRUD.java
//java -cp ".;mysql-connector-j-9.6.0.jar" StudentsCRUD

My Sql –

```
CREATE DATABASE college;

USE college;
SHOW table;
CREATE TABLE Students (
   rollNo INT PRIMARY KEY,
   name VARCHAR(50),
   marks INT
);

select * from students;
```

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Experiment No. 5

EmployeeJDBC.java –

```
import java.sql.*;
import java.util.Scanner;

public class EmployeeJDBC {

    static final String URL = "jdbc:mysql://localhost:3306/company";
    static final String USER = "root";
    static final String PASSWORD = "astik45";

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(URL, USER, PASSWORD);

            // INSERT
            String insertSQL = "INSERT INTO Employee VALUES (?, ?, ?)";
            PreparedStatement psInsert = con.prepareStatement(insertSQL);

            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter Employee Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Salary: ");
            double salary = sc.nextDouble();

            psInsert.setInt(1, id);
            psInsert.setString(2, name);
            psInsert.setDouble(3, salary);

            psInsert.executeUpdate();
            System.out.println("Record Inserted Successfully\n");

            // UPDATE
            String updateSQL = "UPDATE Employee SET salary = ? WHERE empId = ?";
            PreparedStatement psUpdate = con.prepareStatement(updateSQL);

            System.out.print("Enter Employee ID to Update Salary: ");
            int uid = sc.nextInt();

            System.out.print("Enter New Salary: ");
            double newSalary = sc.nextDouble();
```

```
            psUpdate.setDouble(1, newSalary);
            psUpdate.setInt(2, uid);

            psUpdate.executeUpdate();
            System.out.println("Salary Updated Successfully\n");

            // CALL STORED PROCEDURE
            CallableStatement cs = con.prepareCall("{call getEmployeeById(?)}");

            System.out.print("Enter Employee ID to Fetch: ");
            int fetchId = sc.nextInt();

            cs.setInt(1, fetchId);

            ResultSet rs = cs.executeQuery();

            while (rs.next()) {
                System.out.println("ID: " + rs.getInt("empId"));
                System.out.println("Name: " + rs.getString("empName"));
                System.out.println("Salary: " + rs.getDouble("salary"));
            }

            con.close();

        } catch (Exception e) {
            e.printStackTrace();
        }

        sc.close();
    }
}
```

My Sql code –

```
create database company;

use company;

CREATE TABLE Employee (
   empId INT PRIMARY KEY,
   empName VARCHAR(40),
   salary DOUBLE
);

DELIMITER //
CREATE PROCEDURE getEmployeeById(IN eid INT)
BEGIN
   SELECT * FROM Employee WHERE empId = eid;
END //
DELIMITER ;
```

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Experiment No. 6

Index.html

```
<!DOCTYPE html>

<html>
<head>
   <title>Greeting App</title>
</head>
<body>
   <h2>Enter Your Name</h2>

   <form action="greet" method="post">
      <input type="text" name="username" placeholder="Enter name" required>
      <button type="submit">Submit</button>
   </form>

</body>
</html>
```

GreetingServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class GreetingServlet extends HttpServlet {

   protected void doPost(HttpServletRequest request,
                         HttpServletResponse response)
         throws ServletException, IOException {

      String name = request.getParameter("username");

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();

      out.println("<html><body>");
      out.println("<h2>Hello, " + name + "! Welcome to Servlet Application.</h2>");
      out.println("</body></html>");
   }
}
```

Web.xml

```
<web-app>

   <servlet>
      <servlet-name>GreetingServlet</servlet-name>
      <servlet-class>GreetingServlet</servlet-class>
   </servlet>
```

```
  <servlet-mapping>
    <servlet-name>GreetingServlet</servlet-name>
    <url-pattern>/greet</url-pattern>
  </servlet-mapping>

</web-app>
```

Subject – FSD

Experiment No. 7

Index.html

```
<!DOCTYPE html>
<html>
<head>
<title>Calculator Servlet</title>
</head>
<body>

<h2>Simple Calculator</h2>
<form action="calculate" method="post">

Enter Number 1:
<input type="text" name="num1"><br><br>

Enter Number 2:
<input type="text" name="num2"><br><br>
<input type="submit" name="operation" value="Add">
<input type="submit" name="operation" value="Subtract">

</form>
</body>
</html>
```

Web.xml

```
<web-app>
<servlet>
<servlet-name>CalculateServlet</servlet-name>
<servlet-class>CalculateServlet</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>CalculateServlet</servlet-name>
<url-pattern>/calculate</url-pattern>
</servlet-mapping>
</web-app>
```

Servlet Mapping

```
  <servlet-mapping>
    <servlet-name>GreetingServlet</servlet-name>
    <url-pattern>/greet</url-pattern>
  </servlet-mapping>
  <!-- Welcome file -->
  <welcome-file-list>
    <welcome-file>index.html</welcome-file>
  </welcome-file-list>
</web-app>
```

Subject – FSD

Experiment No. 8

Index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Employee Web Application</title>
</head>
<body>
<h1>Welcome to Employee Web Application</h1>
<p>This is the Home Page</p>
<a href="login.html">Go to Login Page</a>
</body>
</html>
```

Login.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Employee Login</title>
</head>
<body>
<h2>Employee Login Form</h2>
<form>
<label>Employee ID:</label>
<input type="text" name="empid"><br><br>
<label>Password:</label>
<input type="password" name="password"><br><br>
<input type="submit" value="Login">
<input type="reset" value="Clear">
</form>
</body>
</html>
```

Servlet.java

```
java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class LoginServlet extends HttpServlet {
   public void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String id = request.getParameter("empid");
String pass = request.getParameter("password");
```

```
        if(id.equals("admin") && pass.equals("123"))
{out.println("<h1>Login Successful</h1>");
        } else {
            out.println("<h1>Login Failed</h1>");
        }
    }
}
```

Subject – FSD

Experiment No. 9

Index.html

```
<html>
<head>
<title>Cookie Example</title>
</head>
<body>
<h2>Enter Your Name</h2>
<form action="CookieServlet" method="post">
Name: <input type="text" name="username">
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Cookie servlet.java

```
package com.example;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class CookieServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request, HttpServletResponse response)
          throws ServletException, IOException {
  response.setContentType("text/html");
       PrintWriter out = response.getWriter();

       String name = request.getParameter("username");
  Cookie cookie = new Cookie("username", name);
       cookie.setMaxAge(60*60);   response.addCookie(cookie);
out.println("<h2>Cookie Created Successfully</h2>");
       out.println("<a href='ReadCookie'>Read Cookie</a>");
    }
}
```

Read Cookoie.java

```
          }package com.example;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class ReadCookie extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
          throws ServletException, IOException {

       response.setContentType("text/html");
       PrintWriter out = response.getWriter();
      Cookie cookies[] = request.getCookies();  if(cookies != null) {
   for(Cookie c : cookies) {
      if(c.getName().equals("username")) {
```

```
                out.println("<h2>Welcome " + c.getValue() + "</h2>");
            }
        } else {
    out.println("No cookies found");

        }
    }
}
```

Subject – FSD

Experiment No. 10

Index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>JSP Greeting Form</title>
</head>
<body>
   <h2>Enter Your Name</h2>
   <form action="greet.jsp" method="post">
      Name:
      <input type="text" name="username" required>
      <br><br>
      <input type="submit" value="Submit">
   </form>
</body>
</html>
```

greet.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
   pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
   <title>Greeting Page</title>
</head>
<body>

<%
   String name = request.getParameter("username");
%>

<h2>Hello <%= name %>, Welcome!</h2>

<br><br>
<a href="index.html">Go Back</a>

</body>
</html>
```

Subject – FSD

Experiment No. 11

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
   <modelVersion>4.0.0</modelVersion>
   <groupId>com.example</groupId>
   <artifactId>HibernateStudentApp</artifactId>
   <version>1.0-SNAPSHOT</version>
   <dependencies>
      <!-- Hibernate Core -->
      <dependency>
         <groupId>org.hibernate</groupId>
         <artifactId>hibernate-core</artifactId>
         <version>5.6.15.Final</version>
      </dependency>
      <!-- MySQL Connector -->
      <dependency>
         <groupId>mysql</groupId>
         <artifactId>mysql-connector-java</artifactId>
         <version>8.0.33</version>
      </dependency>
      <!-- JPA API -->
      <dependency>
         <groupId>javax.persistence</groupId>
         <artifactId>javax.persistence-api</artifactId>
         <version>2.2</version>
      </dependency>
   </dependencies>
</project>
```

hibernate.cfg.xml

```
<!DOCTYPE hibernate-configuration PUBLIC
 "-//Hibernate/Hibernate Configuration DTD 3.0//EN"
 "http://hibernate.sourceforge.net/hibernate-configuration-3.0.dtd">
<hibernate-configuration>
 <session-factory>
  <!-- Database connection -->
  <property name="hibernate.connection.driver_class">com.mysql.cj.jdbc.Driver</property>
  <property name="hibernate.connection.url">jdbc:mysql://localhost:3306/testdb</property>
  <property name="hibernate.connection.username">root</property>
  <property name="hibernate.connection.password">#Sbg@9356171790%</property>
  <!-- Dialect -->
  <property name="hibernate.dialect">org.hibernate.dialect.MySQL8Dialect</property>
  <!-- Show SQL -->
  <property name="hibernate.show_sql">true</property>
  <!-- Auto create/update table -->
  <property name="hibernate.hbm2ddl.auto">update</property>
```

```
  <!-- Mapping -->
  <mapping class="com.example.Student"/>
 </session-factory>
</hibernate-configuration>
```

```
package com.example;

import org.hibernate.SessionFactory;
import org.hibernate.cfg.Configuration;

public class HibernateUtil {
   private static final SessionFactory sessionFactory =
         new Configuration().configure().buildSessionFactory();

   public static SessionFactory getSessionFactory() {
      return sessionFactory;
   }
}
```

Student.java

```
package com.example;

import javax.persistence.*;

@Entity
public class Student {

   @Id
   @GeneratedValue(strategy = GenerationType.IDENTITY)
   private int id;

   private String name;
   private int marks;

   public Student() {}

   public Student(String name, int marks) {
      this.name = name;
      this.marks = marks;
   }

   // Getters
   public int getId() { return id; }
   public String getName() { return name; }
   public int getMarks() { return marks; }

   // Setters
   public void setName(String name) { this.name = name; }
   public void setMarks(int marks) { this.marks = marks; }
}
```

MainApp.java

```java
package com.example;

import org.hibernate.Session;
import org.hibernate.Transaction;
import org.hibernate.query.Query;
import java.util.List;
import java.util.Scanner;

public class MainApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("\n===== STUDENT MANAGEMENT USING HIBERNATE =====");
            System.out.println("1. Insert Student");
            System.out.println("2. View All Students");
            System.out.println("3. Update Student Marks");
            System.out.println("4. Delete Student");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1: insertStudent(sc); break;
                case 2: viewStudents(); break;
                case 3: updateStudent(sc); break;
                case 4: deleteStudent(sc); break;
                case 5: System.out.println("Exiting program..."); break;
                default: System.out.println("Invalid choice! Please try again.");
            }
        } while (choice != 5);
        sc.close();
    }

    // 1. Insert Student
    public static void insertStudent(Scanner sc) {
        Session session = HibernateUtil.getSessionFactory().openSession();
        Transaction tx = session.beginTransaction();

        System.out.print("Enter Student Name: ");
        sc.nextLine(); // consume newline
        String name = sc.nextLine();

        System.out.print("Enter Student Marks: ");
        int marks = sc.nextInt();

        Student s = new Student(name, marks);
        session.save(s);
        tx.commit();
        session.close();
        System.out.println("✅ Student Inserted Successfully!");
    }
```

```
// 2. View All Students
public static void viewStudents() {
    Session session = HibernateUtil.getSessionFactory().openSession();
    Query<Student> query = session.createQuery("from Student", Student.class);
    List<Student> students = query.list();

    System.out.println("\n===== STUDENT RECORDS =====");
    if (students.isEmpty()) {
        System.out.println("No student records found.");
    } else {
        System.out.printf("%-5s %-20s %-10s\n", "ID", "NAME", "MARKS");
        System.out.println("-------------------------------------");
        for (Student s : students) {
            System.out.printf("%-5d %-20s %-10d\n", s.getId(), s.getName(), s.getMarks());
        }
    }
    session.close();
}

// 3. Update Student Marks
public static void updateStudent(Scanner sc) {
    Session session = HibernateUtil.getSessionFactory().openSession();
    Transaction tx = session.beginTransaction();

    System.out.print("Enter Student ID to update: ");
    int id = sc.nextInt();

    Student s = session.get(Student.class, id);
    if (s != null) {
        System.out.print("Enter New Marks: ");
        int newMarks = sc.nextInt();
        s.setMarks(newMarks);
        session.update(s);
        tx.commit();
        System.out.println("✅ Student Marks Updated Successfully!");
    } else {
        System.out.println("❌ Student not found!");
    }
    session.close();
}

// 4. Delete Student
public static void deleteStudent(Scanner sc) {
    Session session = HibernateUtil.getSessionFactory().openSession();
    Transaction tx = session.beginTransaction();

    System.out.print("Enter Student ID to delete: ");
    int id = sc.nextInt();

    Student s = session.get(Student.class, id);
    if (s != null) {
        session.delete(s);
```

```
            tx.commit();
            System.out.println("✅ Student Deleted Successfully!");
        } else {
            System.out.println("❌ Student not found!");
        }
        session.close();
    }
}
```

Subject – FSD

Experiment No. 12

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">

   <modelVersion>4.0.0</modelVersion>
   <groupId>com.example</groupId>
   <artifactId>SpringCoreDemo</artifactId>
   <version>1.0-SNAPSHOT</version>
   <packaging>jar</packaging>

   <properties>
      <maven.compiler.source>17</maven.compiler.source>
      <maven.compiler.target>17</maven.compiler.target>
   </properties>

   <dependencies>
      <!-- Spring Context -->
      <dependency>
         <groupId>org.springframework</groupId>
         <artifactId>spring-context</artifactId>
         <version>5.3.30</version>
      </dependency>
   </dependencies>
</project>
```

Course.java

```
package com.example;

public class Course {
   private String courseName;

   public Course() {
      this.courseName = "Spring Framework";
   }

   public String getCourseName() {
      return courseName;
   }
}
```

Teacher.java

```
package com.example;

public class Teacher {
   private int id;
```

```
    private String name;
    private Course course;

    public void setId(int id) {
        this.id = id;
    }

    public void setName(String name) {
        this.name = name;
    }

    public void setCourse(Course course) {
        this.course = course;
    }

    public void display() {
        System.out.println("Teacher ID: " + id);
        System.out.println("Teacher Name: " + name);
        System.out.println("Course: " + course.getCourseName());
    }
}
```

App.java

```
package com.example;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class App {
    public static void main(String[] args) {

        ApplicationContext context =
                new ClassPathXmlApplicationContext("applicationContext.xml");

        Teacher t1 = (Teacher) context.getBean("teacherBean");
        Teacher t2 = (Teacher) context.getBean("teacherBean");

        Course c1 = (Course) context.getBean("courseBean");
        Course c2 = (Course) context.getBean("courseBean");

        System.out.println("---- Teacher Data ----");
        t1.display();

        System.out.println("\n---- Scope Check ----");
        System.out.println("Teacher t1: " + t1);
        System.out.println("Teacher t2: " + t2);

        System.out.println("Course c1: " + c1);
        System.out.println("Course c2: " + c2);
    }
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
      xsi:schemaLocation="
      http://www.springframework.org/schema/beans
      http://www.springframework.org/schema/beans/spring-beans.xsd">

   <!-- Prototype -->
   <bean id="courseBean"
        class="com.example.Course"
        scope="prototype"/>

   <!-- Singleton + Autowiring -->
   <bean id="teacherBean"
        class="com.example.Teacher"
        scope="singleton"
        autowire="byType">

      <property name="id" value="101"/>
      <property name="name" value="Rahul"/>
   </bean>

</beans>
```

Subject – FSD

Experiment No. 13

Product.java

```
package com.example.model;
public class Product {
   private int id;
   private String name;
   private double price;

   public Product() {
   }
   public int getId() {
      return id;
   }
   public void setId(int id) {
      this.id = id;
   }   public String getName() {
      return name;
   }
   public void setName(String name) {
      this.name = name;
   }   public double getPrice() {
      return price;
   }
   public void setPrice(double price) {
      this.price = price;
   }
}
```

ProductController.java

```
package com.example.controller;

import com.example.model.Product;
import com.example.service.ProductService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;

@Controller
public class ProductController {

   @Autowired
   private ProductService productService;

   @RequestMapping(value = "/productForm", method = RequestMethod.GET)
   public String showForm(Model model) {
      model.addAttribute("product", new Product());
      return "productForm";
   }

   @RequestMapping(value = "/saveProduct", method = RequestMethod.POST)
```

```
    public String saveProduct(@ModelAttribute("product") Product product, Model model) {
        productService.saveProduct(product);
        model.addAttribute("savedProduct", product);
        return "success";
    }
}
```

ProductDAO.java

```
package com.example.dao;

import com.example.model.Product;
import org.springframework.jdbc.core.JdbcTemplate;

public class ProductDAO {

    private JdbcTemplate jdbcTemplate;

    public void setJdbcTemplate(JdbcTemplate jdbcTemplate) {
        this.jdbcTemplate = jdbcTemplate;
    }

    public int saveProduct(Product product) {
        String sql = "INSERT INTO product (id, name, price) VALUES (?, ?, ?)";
        return jdbcTemplate.update(sql, product.getId(), product.getName(), product.getPrice());
    }
}
```

LoggingAspect.java

```
package com.example.aspect;

import org.aspectj.lang.annotation.Aspect;
import org.aspectj.lang.annotation.Before;

@Aspect
public class LoggingAspect {

    @Before("execution(* com.example.service.ProductService.saveProduct(..))")
    public void logBeforeSave() {
        System.out.println("AOP Log: Product is about to be saved...");
    }
}
```

productForm.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<html>
<head>
    <title>Add Product</title>
</head>
<body>
    <h2>Add New Product</h2>

    <form action="saveProduct" method="post">
        Product ID: <input type="text" name="id"><br><br>
```

```
    Product Name: <input type="text" name="name"><br><br>
    Product Price: <input type="text" name="price"><br><br>
    <input type="submit" value="Save Product">
  </form>
</body>
</html>
```

success.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<html>
<head>
  <title>Success</title>
</head>
<body>
  <h2>Product Saved Successfully!</h2>

  <p><b>ID:</b> ${savedProduct.id}</p>
  <p><b>Name:</b> ${savedProduct.name}</p>
  <p><b>Price:</b> ${savedProduct.price}</p>
</body>
</html>
```

spring-servlet.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:context="http://www.springframework.org/schema/context"
    xmlns:mvc="http://www.springframework.org/schema/mvc"
    xmlns:aop="http://www.springframework.org/schema/aop"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd
    http://www.springframework.org/schema/context
    http://www.springframework.org/schema/context/spring-context.xsd
    http://www.springframework.org/schema/mvc
    http://www.springframework.org/schema/mvc/spring-mvc.xsd
    http://www.springframework.org/schema/aop
    http://www.springframework.org/schema/aop/spring-aop.xsd">

  <context:component-scan base-package="com.example"/>
  <mvc:annotation-driven/>

  <bean class="org.springframework.web.servlet.view.InternalResourceViewResolver">
    <property name="prefix" value="/WEB-INF/views/"/>
    <property name="suffix" value=".jsp"/>
  </bean>
  <bean id="dataSource" class="org.springframework.jdbc.datasource.DriverManagerDataSource">
    <property name="driverClassName" value="com.mysql.cj.jdbc.Driver"/>
    <property name="url" value="jdbc:mysql://localhost:3306/springmvcdb"/>
    <property name="username" value="root"/>
    <property name="password" value="#Sbg@9356171790%"/>
  </bean>
  <bean id="jdbcTemplate" class="org.springframework.jdbc.core.JdbcTemplate">
```

```
    <property name="dataSource" ref="dataSource"/>
  </bean>
  <bean id="productDAO" class="com.example.dao.ProductDAO">
    <property name="jdbcTemplate" ref="jdbcTemplate"/>
  </bean>
  <bean id="productService" class="com.example.service.ProductService">
    <property name="productDAO" ref="productDAO"/>
  </bean>
  <bean id="loggingAspect" class="com.example.aspect.LoggingAspect"/>
  <aop:aspectj-autoproxy/>

</beans>
```

web.xml

```
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
      xsi:schemaLocation="
      http://xmlns.jcp.org/xml/ns/javaee
      http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd"
      version="4.0">

  <display-name>SpringMVCProductApp</display-name>
  <servlet>
    <servlet-name>spring</servlet-name>
    <servlet-class>org.springframework.web.servlet.DispatcherServlet</servlet-class>
    <init-param>
      <param-name>contextConfigLocation</param-name>
      <param-value>/WEB-INF/spring-servlet.xml</param-value>
    </init-param>
    <load-on-startup>1</load-on-startup>
  </servlet>
  <servlet-mapping>
    <servlet-name>spring</servlet-name>
    <url-pattern>/</url-pattern>
  </servlet-mapping>
</web-app>
```

Database:

```
CREATE DATABASE springmvcdb;

USE springmvcdb;

CREATE TABLE product (
  id INT PRIMARY KEY,
  name VARCHAR(100),
  price DOUBLE
);
```

Subject – FSD

Experiment No. 14

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Profile</title>
<link rel="stylesheet" href="style.min.css">
</head>
<body>
<header>
  <h1>My Profile</h1>
  <nav>
    <a href="#profile">Profile</a>
    <a href="#skills">Skills</a>
    <a href="#contact">Contact</a>
  </nav>
</header>

<section id="profile">
  <img src="profile.webp" alt="Profile Image" loading="lazy" width="200" height="200">
  <h2>Astik Pawar</h2>
  <p>Computer Science Engineering student with a passion for Web and AI projects.</p>
  <p>Email: astik@example.com</p>
</section>

<section id="skills" class="skills-container">
  <div class="skill-card">
    <img src="html.webp" alt="HTML" loading="lazy" width="100" height="100">
    <h3>HTML</h3>
  </div>
  <div class="skill-card">
    <img src="css.webp" alt="CSS" loading="lazy" width="100" height="100">
    <h3>CSS</h3>
  </div>
  <div class="skill-card">
    <img src="js.webp" alt="JavaScript" loading="lazy" width="100" height="100">
    <h3>JavaScript</h3>
  </div>
  <div class="skill-card">
    <img src="java.webp" alt="Java" loading="lazy" width="100" height="100">
    <h3>Java</h3>
  </div>
</section>

<footer>
  <p>© 2026 Astik Pawar. All rights reserved.</p>
</footer>
</body>
```

```
</html>
```

Style.css

```
body {
   margin: 0;
   font-family: Arial, sans-serif;
   background: linear-gradient(to right, #6a11cb, #2575fc);
   color: #fff;
}

header {
   padding: 20px;
   text-align: center;
   background: rgba(0, 0, 0, 0.5);
}

header h1 {
   margin: 0;
}

nav a {
   color: #fff;
   text-decoration: none;
   margin: 0 10px;
   font-weight: bold;
}

nav a:hover {
   color: #ff0;
}

section#profile {
   padding: 40px;
   text-align: center;
}

#profile img {
   border-radius: 50%;
   box-shadow: 0 4px 8px rgba(0, 0, 0, 0.3);
   margin-bottom: 20px;
}

.skills-container {
   display: flex;
   flex-wrap: wrap;
   justify-content: center;
   gap: 20px;
   padding: 40px 0;
}

.skill-card {
   background: rgba(255, 255, 255, 0.1);
   padding: 20px;
   border-radius: 10px;
```

```
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.2);
    width: 200px;
    text-align: center;
    transition: transform 0.3s;
}

.skill-card:hover {
    transform: scale(1.1);
}

footer {
    padding: 20px;
    text-align: center;
    background: rgba(0, 0, 0, 0.5);
    margin-top: 20px;
}

@media (max-width: 768px) {
    .skills-container {
        flex-direction: column;
        align-items: center;
    }

    .skill-card {
        width: 80%;
    }
}
```

Subject – FSD

Experiment No. 15

Student.java

```
package com.example;
import java.util.Objects;
public class Student {
   private String name;
   private int marks;
   public Student(String name, int marks) {
      this.name = name;
      this.marks = marks;
   }
   public String getName() { return name; }
   public int getMarks() { return marks; }
   @Override
   public String toString() {
      return name + " (" + marks + ")";
   }
   @Override
   public boolean equals(Object o) {
      if (this == o) return true;
      if (!(o instanceof Student)) return false;
      Student student = (Student) o;
      return marks == student.marks && Objects.equals(name, student.name);
   }
   @Override
   public int hashCode() {
      return Objects.hash(name, marks);
   }
}
```

StudentStreamExample.java

```
package com.example;
import java.util.*;
import java.util.stream.Collectors;
public class StudentStreamExample {
   public static void main(String[] args) {
      List<Student> students = Arrays.asList(
         new Student("Astik", 85),
         new Student("Riya", 72),
         new Student("Amit", 60),
         new Student("Sneha", 85),
         new Student("Vikram", 45),
         new Student("Kavya", 60)
      );

      // 1. Group students by marks
      Map<Integer, List<Student>> groupByMarks = students.stream()
            .collect(Collectors.groupingBy(Student::getMarks));
      System.out.println("Grouped by Marks:");
```

```
        groupByMarks.forEach((marks, list) -> System.out.println(marks + " -> " + list));

        // 2. Average marks
        double averageMarks = students.stream()
                .mapToInt(Student::getMarks)
                .average()
                .orElse(0);
        System.out.println("\nAverage Marks: " + averageMarks);

        // 3. Partition into pass/fail (>=50 passing)
        Map<Boolean, List<Student>> passFail = students.stream()
                .collect(Collectors.partitioningBy(s -> s.getMarks() >= 50));
        System.out.println("\nPassing Students:");
        passFail.get(true).forEach(System.out::println);
        System.out.println("\nFailing Students:");
        passFail.get(false).forEach(System.out::println);
    }
}
```